

## STUDY QUESTIONS

### Industrial Engineering Problems

1. **A Local Bakery:** A local bakery sells fresh bread daily. The baker must decide how many loaves to bake each morning before knowing the exact demand. Each loaf costs 15 TL to produce and sells for 35 TL. Any unsold loaves at the end of the day can be sold as day-old bread for 8 TL. Based on the last 30 days of sales data, the baker wants to determine the optimal production quantity.

**Daily sales data (last 30 days):**

45, 52, 38, 61, 48, 55, 42, 50, 47, 53,  
58, 44, 49, 56, 51, 46, 54, 43, 57, 48,  
60, 41, 52, 49, 55, 47, 59, 50, 46, 53

- a) Calculate the critical ratio for this problem.
  - b) Determine which position (item number) corresponds to the critical ratio you calculated.
  - c) Based on your calculation, how many loaves should the baker produce each day to maximize expected profit?
2. **A flower shop:** A flower shop orders fresh bouquets every morning for sale that day. Each bouquet costs 20 TL to purchase from suppliers and sells for 50 TL. Any unsold bouquets at the end of the day can be sold at a discount for 5 TL. The shop manager has collected sales data from the past 30 days to help determine the optimal order quantity.

**Daily sales data (last 30 days):**

58, 59, 60, 61, 62,  
63, 63, 64, 64, 65,  
65, 66, 66, 67, 67,  
68, 68, 69, 69, 69,  
70, 70, 71, 71, 72,  
72, 73, 74, 75, 76

- a) Calculate the critical ratio for this problem.
  - b) Determine which position (item number) corresponds to the critical ratio you calculated.
  - c) Based on your calculation, how many bouquets should the shop order each day to maximize expected profit?
3. **An electronics retailer:** An electronics retailer is ordering smartwatches for the upcoming holiday season. Each smartwatch costs 800 TL to purchase from the supplier and will be sold at a retail price of 1,200 TL. Any unsold smartwatches after the holiday season will be sold at a clearance price of 500 TL. Historical data shows that demand follows a continuous uniform distribution between 100 and 300 units.

**Hint:** For a uniform distribution between  $a$  and  $b$ , the cumulative distribution function is:

$$F(x) = \frac{x - a}{b - a} \quad \text{for } a \leq x \leq b$$

- a) Write down the critical ratio formula and calculate the critical ratio for this problem.
- b) Calculate the optimal order quantity.
- c) If the purchase cost increases to 900 TL while other prices remain the same, would the optimal order quantity increase or decrease? Calculate the new optimal quantity to verify your answer.

4. **Emergency Services Coverage:** A city needs to place emergency response teams to cover all 8 neighborhoods ( $U = \{1, 2, 3, 4, 5, 6, 7, 8\}$ ). Each potential station location can cover multiple neighborhoods, but has different setup costs. The city wants to minimize the total cost while ensuring all neighborhoods are covered.

Available sets (potential station locations) and their coverage:

| Set   | Covers Neighborhoods | Cost (TL) |
|-------|----------------------|-----------|
| $S_1$ | $\{1, 2, 3\}$        | 12,000    |
| $S_2$ | $\{2, 4, 5\}$        | 15,000    |
| $S_3$ | $\{3, 5, 6, 7\}$     | 20,000    |
| $S_4$ | $\{4, 7, 8\}$        | 10,000    |
| $S_5$ | $\{1, 6, 8\}$        | 18,000    |

Apply the Greedy Heuristic algorithm to find a solution to this set covering problem. What is the total cost of the solution found by the greedy heuristic? For each iteration, show:

- Which elements are currently uncovered (the set  $U'$ )?
  - Calculate the efficiency for each remaining set.
  - Which set is selected (the one with highest efficiency)?
  - Update the covered elements.
  - Continue until all elements are covered.
5. **Wi-Fi Router Placement:** A university campus has 10 buildings that need Wi-Fi coverage ( $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$ ). The IT department has identified 6 possible locations for Wi-Fi routers, where each router can cover multiple buildings. Each location has a different installation and maintenance cost. Find the minimum cost solution using the greedy heuristic.

Available sets (router locations) and their coverage:

| Set   | Covers Buildings     | Cost (TL) |
|-------|----------------------|-----------|
| $S_1$ | $\{1, 2, 3, 4\}$     | 8,000     |
| $S_2$ | $\{3, 4, 5, 6, 7\}$  | 10,000    |
| $S_3$ | $\{5, 6, 8, 9\}$     | 9,000     |
| $S_4$ | $\{1, 2, 8, 9, 10\}$ | 12,000    |
| $S_5$ | $\{7, 9, 10\}$       | 6,000     |
| $S_6$ | $\{2, 4, 6, 8, 10\}$ | 11,000    |

Apply the Greedy Heuristic algorithm step by step. What sets are selected, and what is the total cost? Show all your work for each step. For each iteration:

- List the uncovered elements.
  - Calculate the efficiency for each available set.
  - Select the set with the highest efficiency.
  - Update which elements have been covered.
6. **Security Camera Installation:** A shopping mall wants to install security cameras to monitor all 6 entrances and corridors ( $U = \{A, B, C, D, E, F\}$ ). Each camera location can monitor multiple areas, but has different installation costs. The mall management wants to find a cost-effective solution.

Available sets (camera locations) and their coverage:

| Set   | Covers Areas  | Cost (TL) |
|-------|---------------|-----------|
| $S_1$ | $\{A, B, C\}$ | 6,000     |
| $S_2$ | $\{B, D, E\}$ | 8,000     |
| $S_3$ | $\{A, C, D\}$ | 7,000     |
| $S_4$ | $\{D, E, F\}$ | 9,000     |
| $S_5$ | $\{C, F\}$    | 5,000     |

Use the Greedy Heuristic to find a solution. What is the final set of camera locations selected, and what is the total installation cost? Show your work clearly:

- For each iteration, show which areas remain uncovered.
- Calculate efficiency for each set that can cover at least one uncovered area.
- Identify which set is chosen in each iteration.
- Track the total cost as you add sets to your solution.

7. **A delivery company:** A delivery company needs to load packages onto a truck with limited weight capacity. Each package has a value (priority) and weight. Use Dynamic Programming to find the optimal selection.

- Items:** 4 packages labeled A, B, C, D
- Weights:**  $w_A = 2$ ,  $w_B = 3$ ,  $w_C = 4$ ,  $w_D = 5$  kg
- Values:**  $p_A = 10$ ,  $p_B = 15$ ,  $p_C = 20$ ,  $p_D = 25$  TL
- Knapsack capacity:**  $W = 8$  kg

$$V(i, w) = \begin{cases} V(i-1, w) & \text{if } w_i > w \\ \max\{V(i-1, w), p_i + V(i-1, w - w_i)\} & \text{if } w_i \leq w \end{cases}$$

where  $V(i, w)$  = maximum value using first  $i$  items with capacity  $w$ .

- Fill in the complete DP table below. Show your work for at least 3 cells to demonstrate understanding of the recursive relation.
- What is the maximum total value that can be achieved? (This is  $V(4, 8)$ )
- Which packages should be selected to achieve this maximum value?

**DP Table to Complete:**

| Capacity (w) | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
|--------------|---|---|---|---|---|---|---|---|---|
| Item-null    | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| Item A (w=2) |   |   |   |   |   |   |   |   |   |
| Item B (w=3) |   |   |   |   |   |   |   |   |   |
| Item C (w=4) |   |   |   |   |   |   |   |   |   |
| Item D (w=5) |   |   |   |   |   |   |   |   |   |

8. **A traveler:** A traveler is packing valuable items for a trip. The following partially completed DP table shows the solution process.

- Items:** 3 items labeled X, Y, Z
- Weights:**  $w_X = 1$ ,  $w_Y = 2$ ,  $w_Z = 3$  kg
- Values:**  $p_X = 6$ ,  $p_Y = 10$ ,  $p_Z = 12$  TL
- Knapsack capacity:**  $W = 5$  kg

**Partially Completed DP Table:**

| Capacity (w) | 0 | 1 | 2  | 3  | 4 | 5 |
|--------------|---|---|----|----|---|---|
| Item-null    | 0 | 0 | 0  | 0  | 0 | 0 |
| Item X (w=1) | 0 | 6 | 6  | 6  | 6 | 6 |
| Item Y (w=2) | 0 | 6 | ?  | 16 | ? | ? |
| Item Z (w=3) | 0 | 6 | 10 | ?  | ? | ? |

- Calculate  $V(2, 2)$  (Item Y, capacity 2). Show your calculation using the recursive relation.
- Calculate  $V(2, 4)$  (Item Y, capacity 4). Show your work.
- Calculate  $V(3, 3)$  (Item Z, capacity 3). Show your work.
- What is the final optimal value  $V(3, 5)$ ?

9. **For each scenario below, answer the question and explain your reasoning based on critical path logic. No calculations are required.**

**Scenario A:**

A software development project has the following critical path identified:

Requirements Analysis → Backend Development → Integration → Testing

There are also non-critical tasks including UI Design, Documentation, and Database Optimization.

The project manager has a limited budget and can choose to invest in speeding up exactly ONE task. The CEO suggests investing in "Database Optimization" because it's currently taking the longest time among all non-critical tasks.

**Question:** Will this investment reduce the overall project completion time? Why or why not?

**Scenario B:**

A manufacturing project has been analyzed, and the optimal solution shows:

- Task M starts at time 10 and takes 5 days
- Task N starts at time 20 and takes 8 days
- The precedence constraint is  $s_N \geq s_M + d_M$

**Question:** Is the constraint between M and N tight? What does this tell you about whether task M is on the critical path? If the company reduces task M's duration by 3 days, will this affect the overall project completion time?

**Scenario C:**

An event planning company is organizing a conference. They have identified that the critical path consists of: Venue Booking → Stage Setup → Audio/Visual Installation → Rehearsal.

The marketing team (not on the critical path) asks for additional budget to speed up their promotional campaign, arguing that faster marketing will bring more attendees and make the event more successful.

**Question:** From a project completion time perspective, should the company allocate budget to speed up the marketing task? Explain your answer.

**Scenario D:**

A research project has two parallel paths:

- Path 1: Literature Review → Data Collection → Analysis → Report Writing (40 days total)
- Path 2: Equipment Setup → Experiments → Data Processing → Report Writing (40 days total)

Both paths end at "Report Writing" and both are critical paths with 40 days duration.

**Question:** The research team wants to finish the project 5 days earlier. They can either: (1) Reduce "Data Collection" on Path 1 by 5 days, OR (2) Reduce both "Data Collection" on Path 1 by 3 days AND "Experiments" on Path 2 by 3 days. Which option will successfully reduce the project completion time? Explain.

10. **Multiple Investment Scenarios:** An automotive company is launching a new car model. The critical path analysis has been completed, and the critical path is:

Design → Prototype → Safety Testing → Production → Marketing Launch

Total project duration: 18 months

Investment Options: The company has received several proposals from different departments:

- i) Design team: "Give us extra budget and we'll finish Design 2 months faster." (Design is on critical path)

- ii) Supplier Relations: "Invest in us to speed up Parts Procurement by 3 months." (Parts Procurement is NOT on critical path)
- iii) Safety Testing: "With more equipment, we can reduce Safety Testing by 1 month." (Safety Testing is on critical path)
- iv) Market Research: "Additional resources will let us complete Market Research 2 months faster." (Market Research is NOT on critical path)

**Questions:**

- a) If the company can only invest in ONE proposal and their goal is to reduce the overall project completion time, which proposal(s) should they consider? Which proposals would NOT achieve this goal?
- b) The CFO says: "Proposals 2 and 4 are cheaper than proposals 1 and 3. Let's invest in the cheaper options to save money while still reducing project time." Is the CFO's reasoning correct? Explain.
- c) The CEO approves both proposals 1 and 3 (reducing Design by 2 months and Safety Testing by 1 month). What will be the new project completion time? Explain your reasoning.
- d) After implementing proposals 1 and 3, a project team member suggests: "Now that we've reduced the critical path, maybe Parts Procurement has become critical and we should invest in proposal 2." Under what conditions could this suggestion be correct?